

2.1 Levels of Organisation

2.2 Symmetry

2.3 Coelom

2.4 Animal tissues

2.1 LEVELS OF ORGANISATION

Living world includes a large number of acellular and multicellular organisms with vast differences in their structure and form. In spite of the differences in the structure and form of different animals, there are fundamental features common to various individuals in relation to the arrangement of cell layers, symmetry and nature of coelom, patterns of digestive, circulatory, reproductive systems, etc.

There are five levels of organisation from the protozoans to the higher metazoans. They are as follows:

Protoplasmic level

Protozoans are the simplest protists. All the basic functions of life such as locomotion, digestion, respiration, excretion, osmoregulation and reproduction are performed by a single cell (The single cell acts as a complete organism) in protozoans. Within the cell, protoplasm is differentiated into cell organelles which carry out different functions. It is called **protoplasmic level of organisation**.

Note: Mention of the protoplasmic level is only to provide better understanding as only metazoans are dealt with in this chapter.

2.1.1 Levels of organisation in the metazoans

Though all the members of the kingdom Metazoa are multicellular, all of them do not exhibit the same pattern of organisation of cells. Cells of the metazoans are not capable of independent existence, and exhibit 'division of labour' at various levels of organisation. Among the metazoans of different kinds, cells may be functionally isolated or grouped together to form tissues, organs or organ-systems and as such structural complexity increases from the lower metazoans to the higher metazoans.

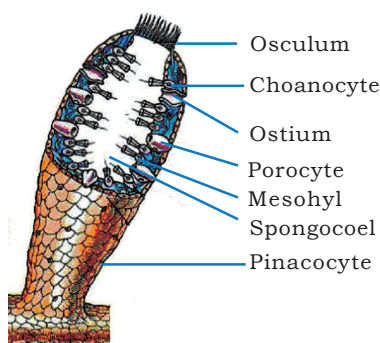


Fig. 2.1 Sponge (body partly cut open) showing cellular level of organisation.

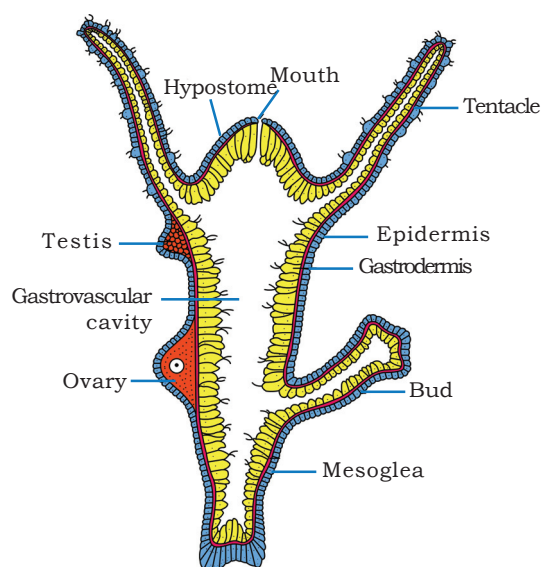


Fig. 2.2 L.S. of Hydra (Showing tissue level of organisation)

The levels of organisation in metazoans are as follows:

I. Cellular level of organisation

It is the lowest level of organisation among the metazoans and is exhibited by the sponges (parazoans). Different types of cells are functionally isolated due to the absence of sensory and nerve cells. The cells are arranged as '**loose cell aggregates**' and do not form tissues. There is division of labour among the cells. The outer layer of body wall is formed by pinacocytes and porocytes (cells that conduct water from the outside into the spongocoel), whereas the inner layer lining the spongocoel is formed by choanocytes. In between the two layers, a non-cellular mesohyl is present.

II. Tissue level of organisation

This is the lowest level of organization among the eumetazoans, exhibited by **diploblastic** animals like the cnidarians. In these animals, the cells which perform the same function are arranged into **tissues**. The cells of a tissue together perform their common function as a highly coordinated unit and this coordination is due to the presence of **nerve cells** and **sensory cells**. **The formation of tissues is the 'first key transition' in the evolution of the animal body plan.** Development of bilateral symmetry is the 'second key transition' and the origin of perivisceral cavity (body cavity) is the 'third key transition'.

III. Organ level of organisation

An aggregation of different kinds of tissues which is specialized for a particular function is called an organ. Organ level of organisation is a further advancement over the tissue level in the evolution of levels of organisation. Organ level of organisation appeared for the first time in the members of the Phylum Platyhelminthes. (Ref: NCERT XI Class biology text book)

IV. Organ-system level of organisation

It is the highest level of organisation among the animals and is exhibited by the triploblastic animals such as the flat worms, nematodes, annelids, arthropods,

molluscs, echinoderms and chordates. In the triploblastic animals, the evolution of '**mesoderm**' resulted in structural complexity. In these animals, the tissues are assembled to form organs and complex organ-systems. Highly specialized sensory and nerve cells bring about a higher level of coordination and integration among the various organ systems to lead an efficient way of life. Organ systems in different groups of animals exhibit various patterns of complexities. For example, the alimentary canal in the members of Platyhelminthes has only a single opening to the exterior, and it serves as both mouth and anus, hence it is called **incomplete gut**. A complete alimentary canal has two openings, the mouth and the anus. Such a gut is seen in the animals ranging from the nematodes to chordates. Similarly, circulatory system may be of two types.

- a) **Open type:** In this type, blood pumped out of the heart flows through open spaces bathing the cells and tissues directly. e.g., members of Arthropoda, Mollusca, Echinodermata and Urochordata.
- b) **Closed type:** In this type, the blood is circulated through a series of vessels of varying diameter (arteries, veins and capillaries.) e.g., annelids, cephalopods, cephalochordates and vertebrates.

During the evolutionary history of animals, there has been a tendency towards increase in the body size, explain Why?

2.1.2 Diploblastic organisation

Animals, in which the cells are arranged in two embryonic layers, the external **ectoderm** and the internal **endoderm**, are called **diploblastic animals**, e.g., cnidarians and ctenophores. Ectoderm gives rise to the **epidermis**, the outer layer of the body wall. Endoderm gives rise to the **gastrodermis**, the tissue that lines the gut cavity. The cells in each tissue layer are functionally interdependent. An undifferentiated layer, present between the ectoderm and endoderm, is the '**mesoglea**'. In some diploblastic animals, cells occur in the mesoglea, but they are always derived either from the ectoderm or from the endoderm.

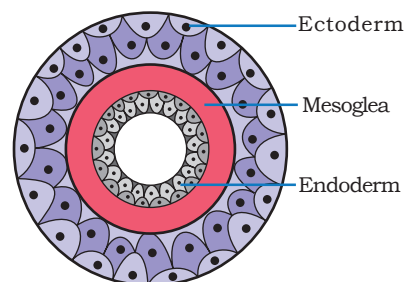


Fig. 2.3 (a) Diploblastic level of organisation

2.1.3 Triploblastic organisation

The animals, in which the developing embryo has a third germinal layer, the **mesoderm**, in between the ectoderm and the endoderm, are called **triploblastic animals**, e.g., flat worms to chordates. Most **triploblastic** animals have an organ-system level of organization. Tissues are organized into organs which form excretory, nervous, digestive, reproductive, circulatory and other systems. Triploblastic animals are usually **bilaterally symmetrical**. (Recall – the taxonomic grade Bilateria is also called Triploblastica).

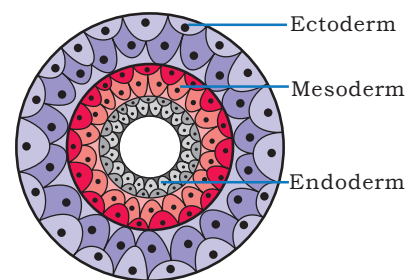


Fig. 2.3 (b) Triploblastic level of organisation